Giant Colony Technique

The Streptomyces culture is inoculated onto the central areas of petriplates containing a nutritious agar medium or they are streaked in a narrow band across the centre of plates. The plates are then incubated until growth and possibly, sporulation have occurred. Strains of micro organisms to be tested for possible sensitivity to the antibiotics (the test organisms) are then streaked from the edges of the plates up to but not touching the *Streptomycete* growth. The plates are further incubated to allow the growth of the test organism. The growth of the test organism has been inhibited by antibiotic in the vicinity of the *Streptomycete* is then measured in millimeters. These *Streptomycetes* that have produced antibiotics with observable microbial inhibition spectrum are retained for further testing (Figure 6.4).

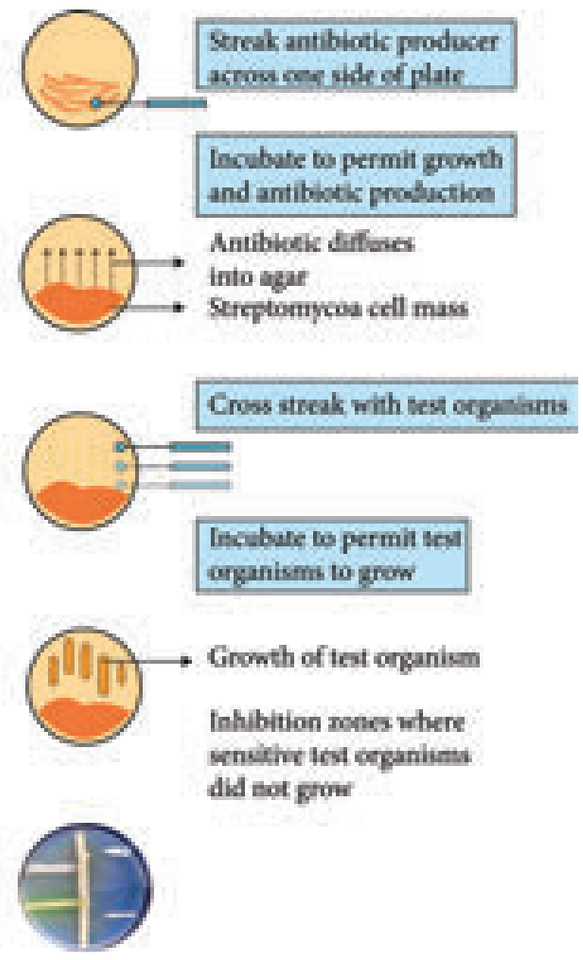


Figure 6.4: Giant Colony Technique

6.3 Strain Improvement

Improvement of the production strain(s) offers the great opportunities for cost reduction without significant capital outlay in industries. Moreover, success in making and keeping a fermentation industry competitive depends greatly on continuous improvement of the production strain(s). Improvement usually resides in increased yields of the desired metabolite. The science and technology of manipulating and improving microbial strains, in order to enhance their metabolic capacities for biotechnological applications, are referred to as strain improvement.

Need for strain improvement

Microbes exist in the nature produce certain compounds of biological interest. However the industrial application of producing those compounds by natural strains is not an economical one so, wild strains are changed by the changing their gene pattern or by regulating their enzymes production. As a result, the specific product is produced in excess.

Knowledge of the function of enzymes, rate limiting steps in pathways, and environmental factors controlling synthesis further helps in designing screening strategies.

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